IMPERIAL COLLEGE LONDON

EE4-27
EE9-CS5-2
EE9-FPN2-08

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected copy

SYSTEMS IDENTIFICATION

Wednesday, 3 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S.A. Evangelou

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1. Consider the electrical circuit depicted in Fig. 1.1.

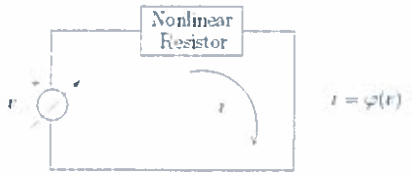


Figure 1.1 Electrical circuit with a nonlinear resistor and a DC power supply.

The model of the resistor is given by a static nonlinear function $i = \varphi(v)$ where v is the voltage and i is the current, and $\varphi(\cdot): [2, 2.5] \mapsto \Re$ is an *unknown* nonlinear function.

An experiment to identify the static model of the resistor is set up in which the voltage v is assumed to be perfectly measurable. A set of M given values of the voltage $v(j), j = 1, \ldots, M$ are imposed by the DC power supply. Correspondingly, a set of M measurements of the current are recorded by a current sensor:

$$m_i(j) = i(j) + \xi(j), j = 1, \ldots, M$$

where $i(j) = \varphi[v(j)]$ and $\xi(\cdot) \sim WN(0, \lambda^2)$.

a) Two approximate models $\mathcal{M}_1$ and $\mathcal{M}_2$ of the unknown function $\varphi(\cdot)$ are considered:

$$\mathcal{M}_1: \widehat{\varphi}_1(v, \theta_1, \theta_2) = \theta_1 e^{-4(v-11/5)^2} + \theta_2 e^{-4(v-12/5)^2}, \quad v \in [2, 2.5]$$

and

$$\mathcal{M}_2: \widehat{\varphi}_2(v, \alpha_1, \alpha_2) = \alpha_1 v + \alpha_2, \quad v \in [2, 2.5]$$

The parameters θ_1, θ_2 of model $\mathcal{M}_1$ and α_1, α_2 of model $\mathcal{M}_2$, respectively, have to be determined using the same set of measurements

$$\Theta = \{(v(j), m_i(j)), j = 1, \ldots, M\}.$$

Devise, if possible, a least-squares method the solution of which provides the optimal (in the least-squares sense) sets of parameters $\theta_1^\circ, \theta_2^\circ$ and $\alpha_1^\circ, \alpha_2^\circ$ of models $\mathcal{M}_1$ and $\mathcal{M}_2$, respectively. [*Hint: Just describe the optimisation problem, but do not attempt to determine the general solution*].

[8 Marks]

b) Determine the general solution of the least-squares method devised in your answer to Question 1-a) and write the specific form that this general solution takes on for model $\mathcal{M}_1$ and for model $\mathcal{M}_2$.

[8 Marks]

c) Consider the following set Θ of 6 input voltages and measurements of current:

$$\Theta = \{(2, 0.03), (2.1, 0.77), (2.2, 0.79), (2.3, 0.9), (2.4, 0.84)(2.5, 0.91)\}$$

Using the solutions obtained in your answer to Question 1-b), compute the optimal parameters $\theta_1^\circ, \theta_2^\circ$ of model $\mathcal{M}_1$ and the optimal parameters $\alpha_1^\circ, \alpha_2^\circ$ of model $\mathcal{M}_2$ and compare the two models. Discuss your findings.

[4 Marks]

2. A stochastic process $w(\cdot)$ is generated by a linear dynamic system according to the block scheme shown in Fig. 2.1.

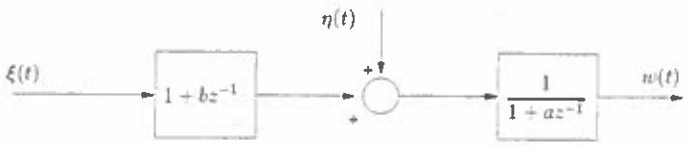


Figure 2.1 Dynamic system generating the stochastic process $w(\cdot)$.

where $a, b \in \Re$, $\eta(\cdot) \sim WN(0, \sigma_\eta^2)$ and $\xi(\cdot) \sim WN(0, \sigma_\xi^2)$ are two mutually independent white noise processes.

a) Suppose that $a = 0$, $b \neq 0$. Show that the stochastic process $w(\cdot)$ is stationary and determine its expected value $\mathbb{E}(w)$, its variance $\text{var}(w)$ and the expression of its correlation function $\gamma_w(\tau)$ for all $\tau \geq 0$.

[4 Marks]

b) Consider the moving-average stochastic process $v(\cdot)$ given by

$$MA(1): \; v(t) = e(t) + ce(t-1)$$

where $c \in \Re$, $c \neq 0$, $|c| < 1$, and $e(\cdot) \sim WN(0, \sigma_e^2)$.

Set $a = 0$, $b = -1/2$, $\sigma_\eta^2 = 1$, $\sigma_\xi^2 = 4$; compute suitable values of c and σ_e^2 such that the stochastic processes $w(\cdot)$ and $v(\cdot)$ have the same spectra, that is

$$\Gamma_w(\omega) = \Gamma_v(\omega), \quad \omega \in [-\pi, \pi]$$

Determine the analytical expression of the spectrum $\Gamma_w(\omega)$ and sketch its diagram in the interval $\omega \in [-\pi, \pi]$.

[7 Marks]

c) Set $a = 1/3$, $b = -1/2$, $\sigma_\eta^2 = 1$, $\sigma_\xi^2 = 4$; determine the spectrum $\Gamma_w'(\omega)$ of the process $w(\cdot)$ and sketch its diagram in the interval $\omega \in [-\pi, \pi]$.

[6 Marks]

d) Compare the spectra $\Gamma_w(\omega)$ and $\Gamma_w'(\omega)$ determined in your answers to Questions 2-b) and 2-c), respectively, and discuss your findings.

[3 Marks]

3. Consider the block scheme shown in Fig. 3.1.

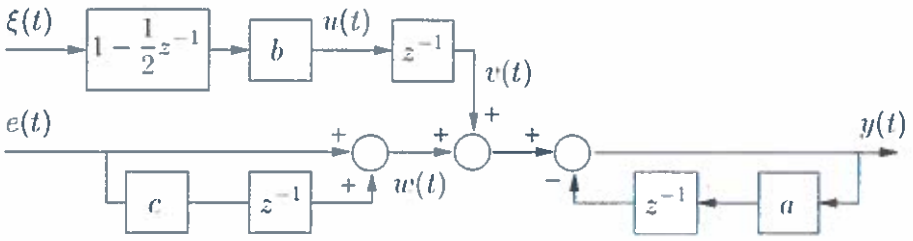


Figure 3.1 Dynamic system generating the stochastic process $y(\cdot)$.

where $e(\cdot) \sim WN(0, \sigma_e^2)$ and $\xi(\cdot) \sim WN(0, \sigma_\xi^2)$ are supposed to be mutually independent. Moreover, $a, b, c \in \Re$, with $a \neq 0$, $|a| < 1$, $b \neq 0$, $c \neq 0$.

a) Write the difference equation expressing $y(t)$ as a function of the inputs $e(t)$ and $\xi(t)$ and show that the stochastic processes $u(\cdot)$ and $y(\cdot)$ are stationary.

[3 Marks]

b) Compute the value of the correlation function of the output process $y(\cdot)$ for $\tau = 1$, that is, $\gamma_y(1) = \mathbb{E}[y(t)y(t-1)]$. Moreover, compute the values for $\tau = 0$ and for $\tau = 1$ of the cross-correlation function between the output process $y(\cdot)$ and the process $u(\cdot)$, that is $\gamma_{yu}(0) = \mathbb{E}[y(t)u(t)]$ and $\gamma_{yu}(1) = \mathbb{E}[y(t)u(t-1)]$.

[7 Marks]

c) Suppose that $u(t)$ is measurable for any time-instant t and consider the prediction model

$$\widehat{\mathscr{M}}(\theta) : \hat{y}(t|t-1) = \alpha y(t-1) + \beta u(t-1), \text{ where } \theta := [\alpha, \beta]^\top .$$

Denote with $\widehat{\theta}(N)$ the least squares estimate of θ based on the set of measurements $\{y(1), y(2), \ldots, y(N), u(1), u(2), \ldots, u(N)\}$ where N is arbitrarily large.

Exploiting the results in your answer to Question 3-b), devise the general formula giving the value $\overline{\theta}$ the estimate $\widehat{\theta}(N)$ approaches for large values of N (that is, $\overline{\theta} = \lim_{N \to \infty} \widehat{\theta}(N)$, a.s.) when the family $\widehat{\mathscr{M}}$ of prediction models is used.

[7 Marks]

d) Set $a = 1/2$, $b = 1/3$, $c = 1/4$, $\sigma_e^2 = 1$, $\sigma_\xi^2 = 9$. Compute the numerical value of $\overline{\theta}$ defined in Question 3-c) and also the variance of the prediction error $\varepsilon(t) = y(t) - \hat{y}(t|t-1)$. Comment on your findings.

[3 Marks]

4. A stochastic process $v(\cdot)$ is generated by a linear dynamic system according to the block scheme shown in Fig. 4.1, where $\xi(\cdot) \sim WN(0,4)$.

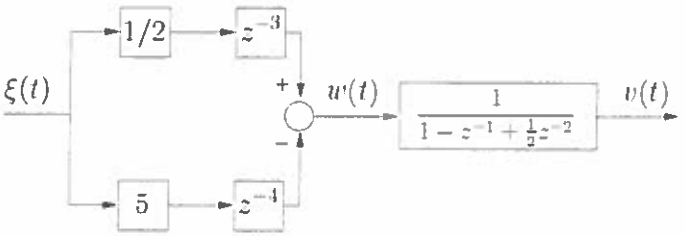


Figure 4.1 Dynamic system generating the stochastic process $v(\cdot)$.

a) Show that the stochastic process $v(\cdot)$ is stationary and compute its variance.

[5 Marks]

b) Determine the transfer function $G(z)$ from the input $\xi(t)$ and the output $v(t)$ and determine the spectral factor in canonical form $\widetilde{G}(z)$.

[4 Marks]

c) Determine the difference equation yielding the optimal one-step ahead prediction $\hat{v}(t+1|t)$ of $v(t+1)$ on the basis of past values $v(t), v(t-1), v(t-2), \ldots$ of the process $v(\cdot)$ and compute the variance of the one-step ahead prediction error $\text{var}[\varepsilon_1(t)] = \text{var}[v(t+1) - \hat{v}(t+1|t)]$.

[4 Marks]

d) Determine the difference equation yielding the optimal two-steps ahead prediction $\hat{v}(t+2|t)$ of $v(t+2)$ on the basis of past values $v(t), v(t-1), v(t-2), \ldots$ of the process $v(\cdot)$ and compute the variance of the two-steps ahead prediction error $\text{var}[\varepsilon_2(t)] = \text{var}[v(t+2) - \hat{v}(t+2|t)]$.

[4 Marks]

e) Compare $\text{var}[\varepsilon_1(t)]$ computed in your answer to Question 4-c) with $\text{var}[\varepsilon_2(t)]$ computed in your answer to Question 4-d) and with the variance of $v(\cdot)$ computed in your answer to Question 4-a). Comment on your findings.

[3 Marks]